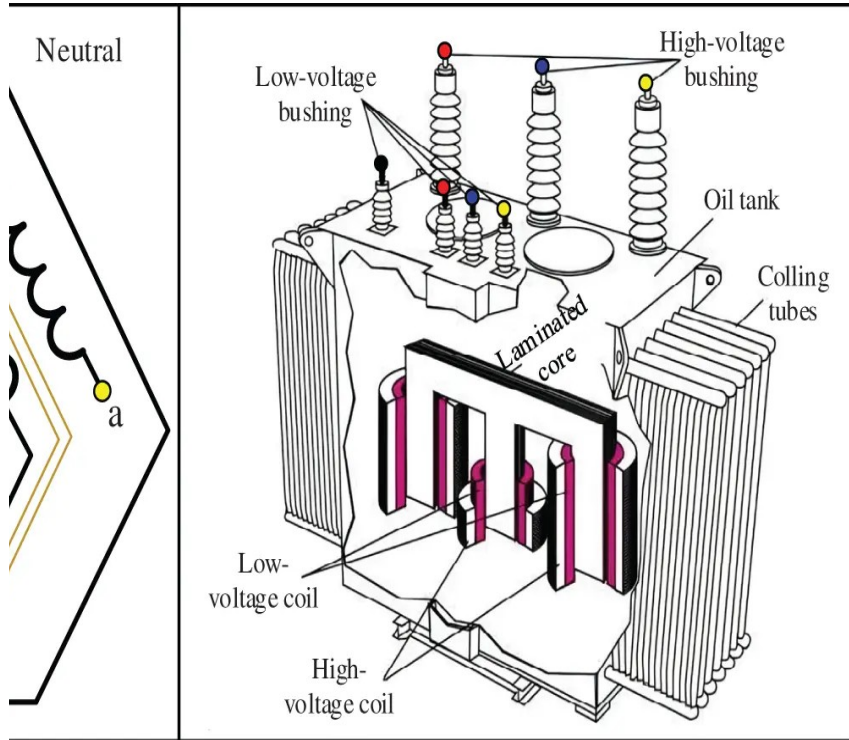


ETTH1

LONG-DURATION FORECASTING OF TRANSFORMER OIL TEMPERATURES

Johannes Aurich |
04 Sep 2026 |
Results Update |

BACKGROUND: OIL IN ELECTRICAL POWER TRANSFORMERS



(Source Link)

What is it used for?

- Electrical insulation to prevent arcs and shorts
- Cooling of windings and core
- Limits moisture and air exposure to prevent aging

Why keep track of oil temperature?

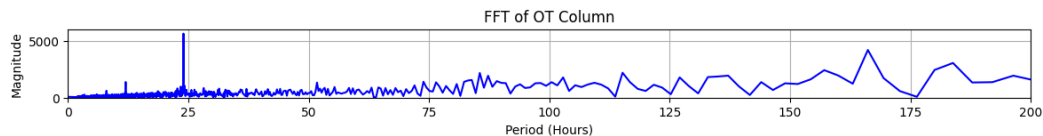
- Optimizing asset performance
- Spikes can indicate faults and overload
- Oil can decompose and fail at too high temperatures

DATA FOR LONG-DURATION FORECASTS

- Electrical substation data from China (pub. 2021)
- <https://github.com/zhouhaoyi/ETDataset>

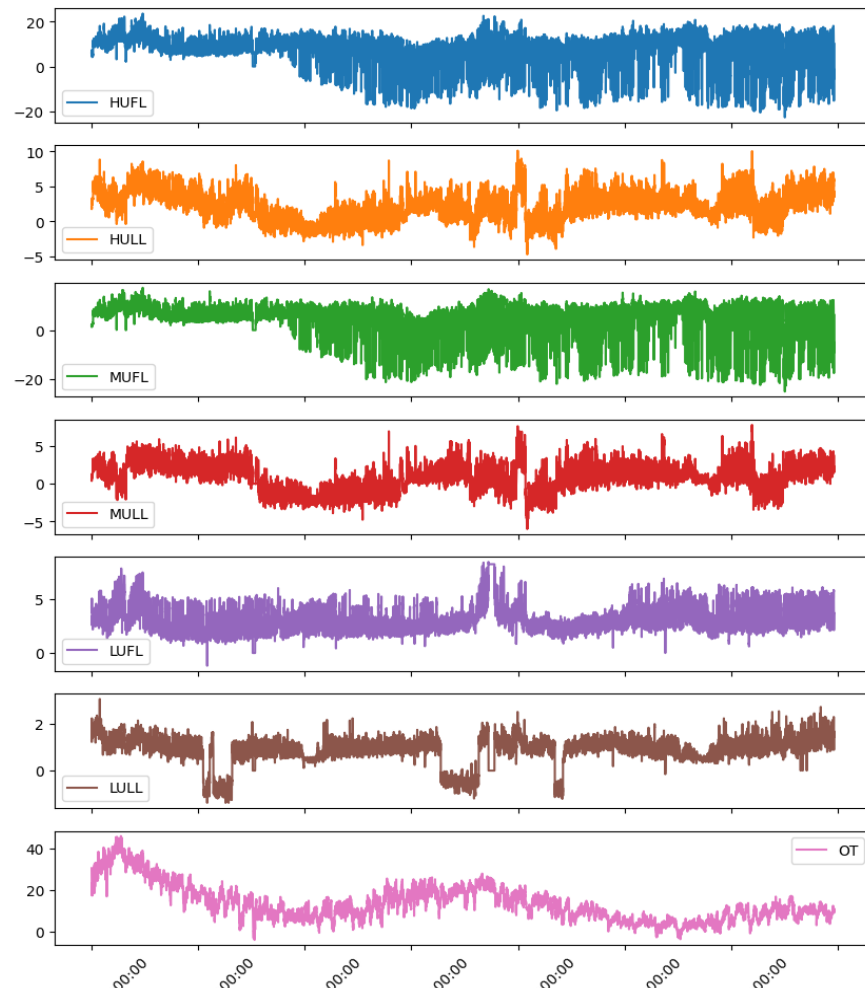
ETTh1 dataset

- 24 months of data | 1h increments
- 7 channels of transformer measurements
- Typ. lookback & forecasting combinations: 96 – 192 – 336 – 720 h
- **Temporal variation** of frequencies, trends, amplitudes, correlations
 - Significant seasonalities, particularly 24h (test split below)



Chosen modeling problem

- 16 h forecast, 96 h lookback horizon/context
- Multivariate input to univariate one-shot forecast of OT time series



SPLITTING AND PREPARATION

Using established splits

- 12 training – 4 validation – 4 test (months) | 60/20/20% split
- Discard the remaining 4 months

Data set definition

- Rolling-window | stride =1 | 64 samples per batch (133 total)

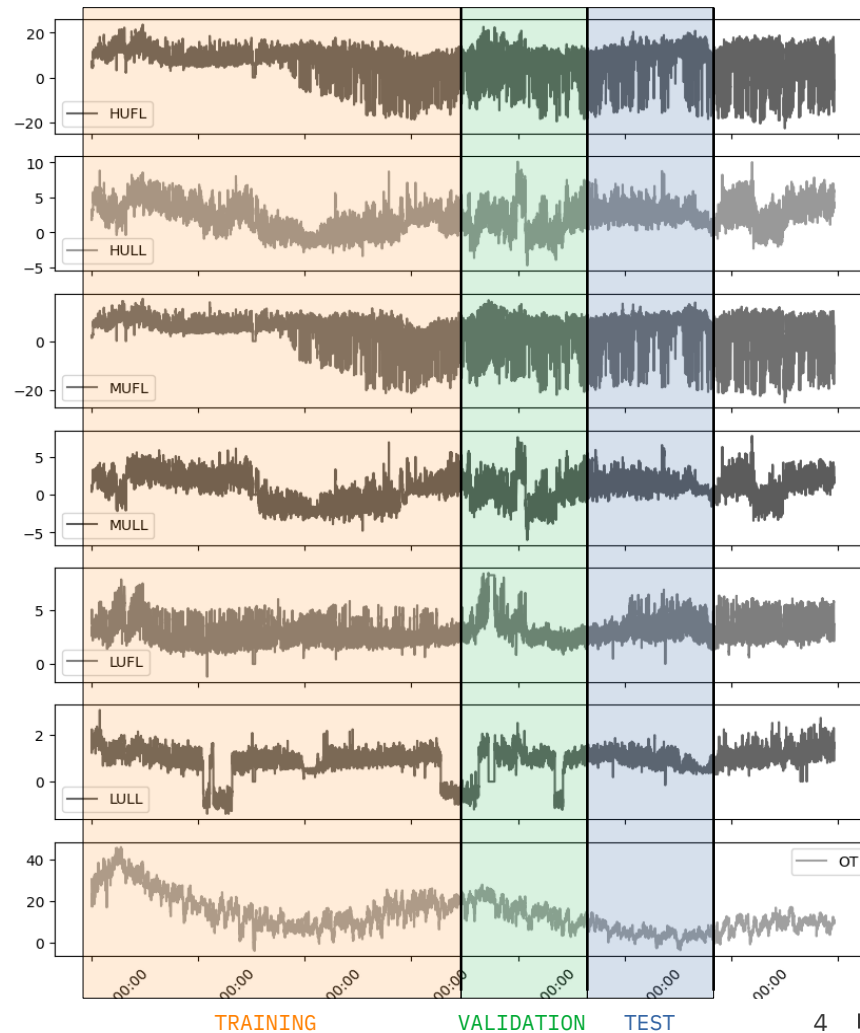
```
training_set = training_split
```

```
validation_set = training_split[-context :] + validation_split
```

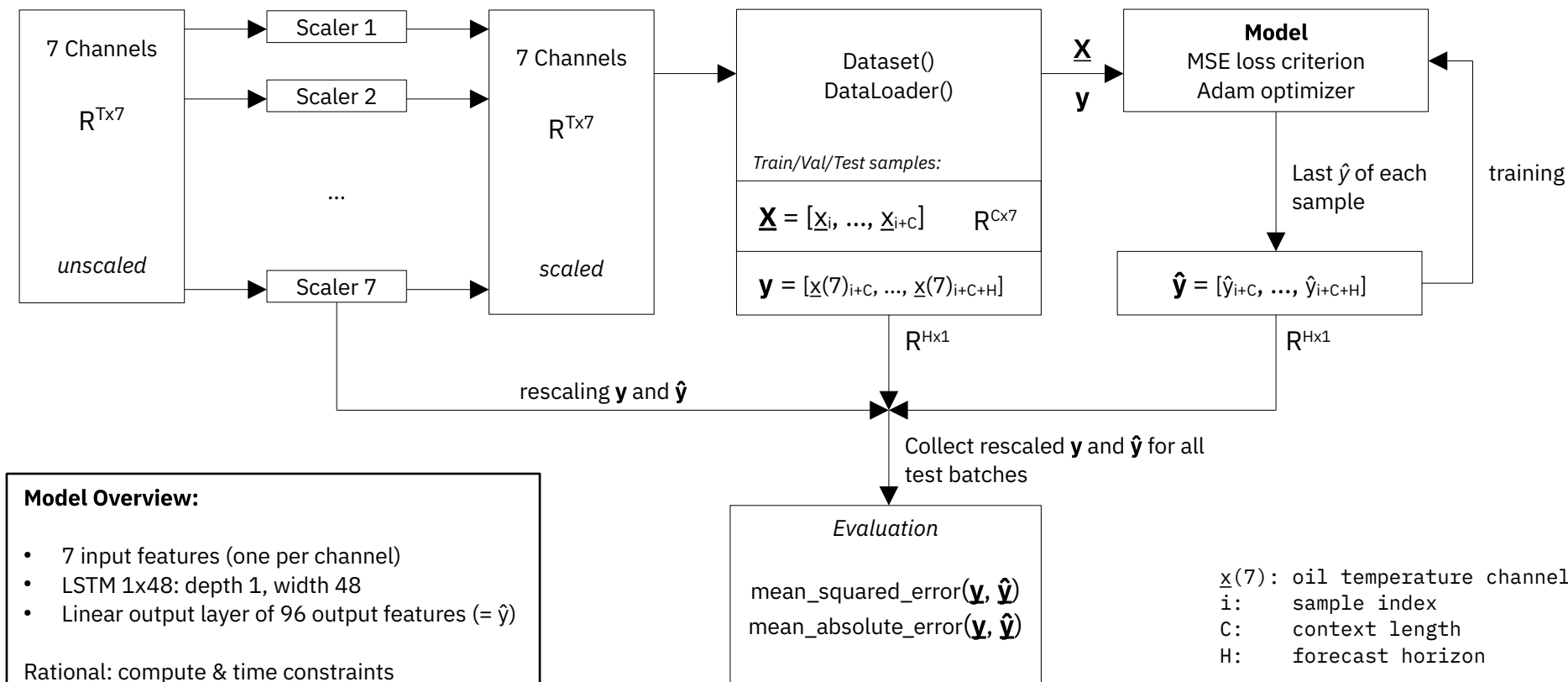
- ```
test_set = validation_split[-context:] + test_split
```

## Scaling

- `MinMaxScaler()`, `feature_range [-1,1]`
- Separate scaler per channel, stored for later `inverse_transform()`
- Per channel: fit on training split, apply to all splits of that channel

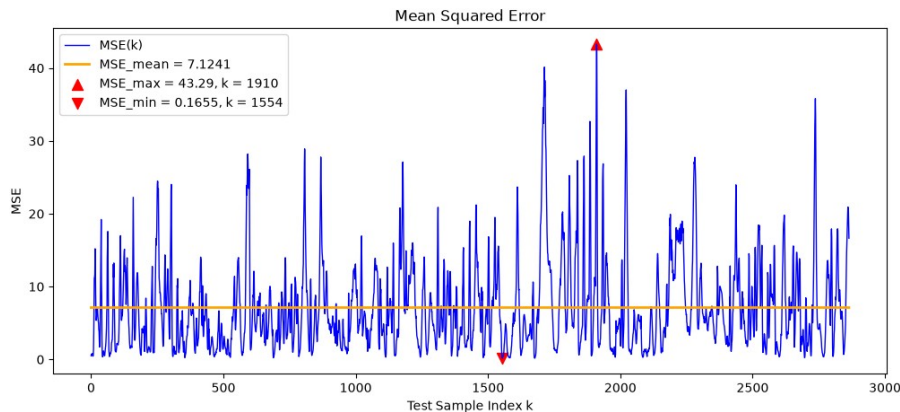


# MODEL OVERVIEW: MULTI-TO-UNIVARIATE SEQUENCE-TO-SEQUENCE



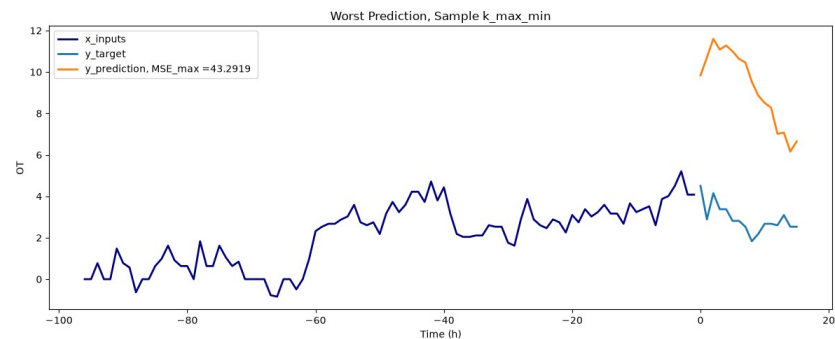
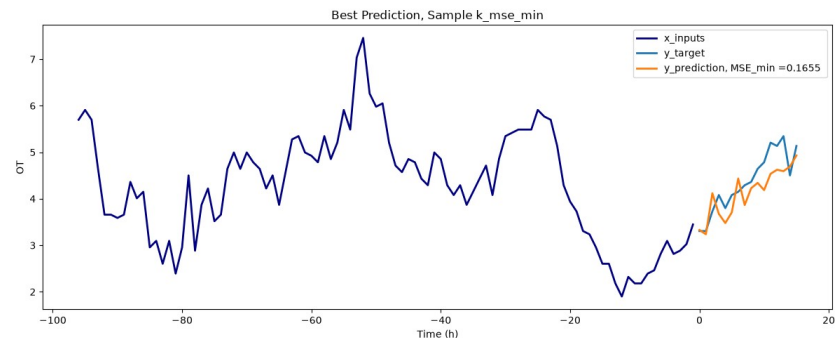
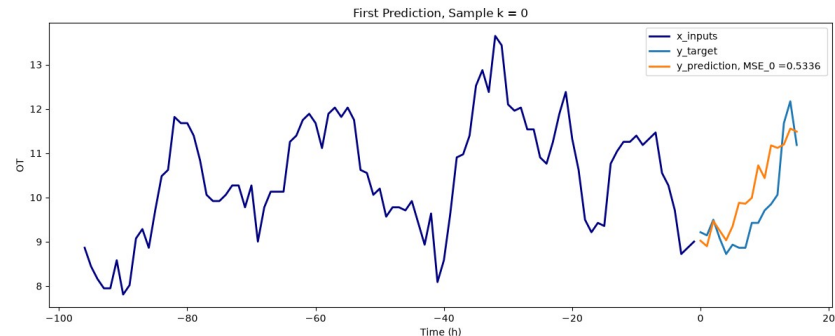
# FORECASTING RESULTS

96H  $\rightarrow$  16H | LSTM 1X48



**Overview** | 11,728 trainable parameters

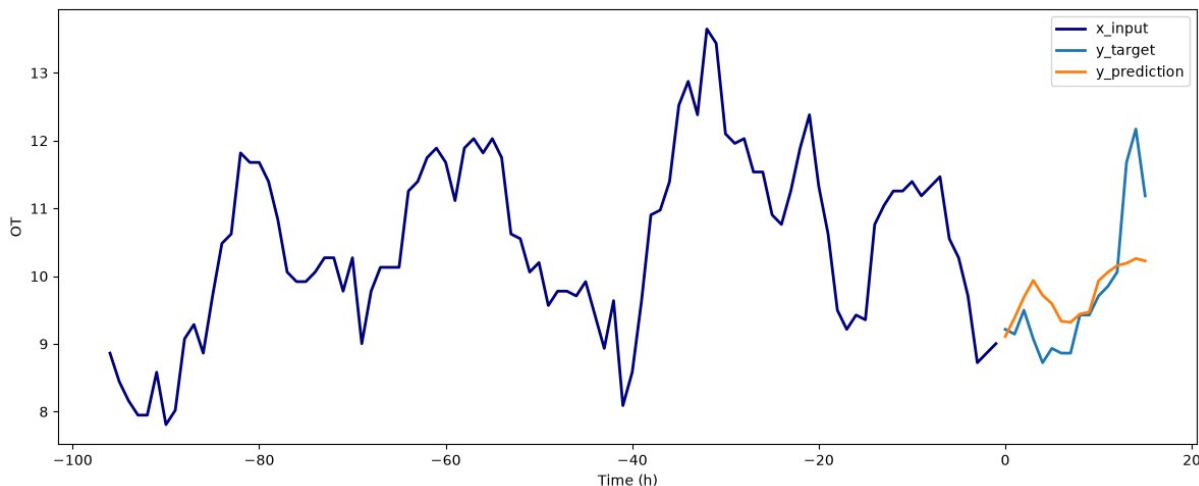
- Test set average MSE: 7.12 | MAE: 2.20
- MSE model performance over the test set fluctuates strongly (see MSE variability above)
- Best test samples achieve MSE  $\approx 0.17$ , showing good reproduction of trend, OT amplitude, and decent fluctuations
- Worst test samples are in the range of MSE  $\approx 40$ , somewhat capturing the general trend but failing on amplitude



# FORECASTING RESULTS

## 96H → 16H | ARIMA BASELINE

- ARIMA = Auto-Regressive Moving Average
- Based on linear differential equations (= may approximate non-linear processes like ETTh1 at best)
- Most useful for short-term predictions



### **auto\_arima()**

*Evaluated for the first test sample only*

- $m = 24$  h seasonality considered
- MSE: 0.60
- MAE: 0.56
- Captures the general trend partially
- Visible approach of the long-term mean, bad reproduction of amplitudes  
→ aligned with typical long-term behavior

- Typical performance of models in the literature:  $MSE, MAE \leq O(1)$  covering transformers (very large) to linear baselines (small)
- → 2022 Paper: “[Are Transformers Effective for Time Series Forecasting?](#)”

# CONCLUSION & NEXT STEPS

## Current models needs further improvement

- LSTM performance fluctuates strongly and requires further measures to stabilize test results (...)
  - Best results are on par with literature values, worst ones significantly underperform; average performance is still insufficient
- ARIMA baseline performs reasonably well on MSE, MAE metrics for the first test sample, but mean MSE/MAE over all test samples is too bad
- More analysis of LSTM network architecture and data handling approach required → hyperparameter optimization
  - Compared to previous experiments: increased lookback-to-forecast horizon ratio helpful, but ideal ratios to be investigated
  - Ideal LSTM configuration: depth and width of network to be investigated, broad networks with at least the output feature range seem beneficial

## Alternative modelling approaches to explore

- ARIMA captures long-term trends better, LSTMs has significant performance variation in amplitudes
  - Try decomposition approaches: *DLinear* → *DLSTM* or *STL-based* = small & simple but limited by primitive seasonality coverage?
  - Use **State Space Models** (inherent trend and dynamics coverage w/o cutoffs like in *FITS*): S4/Mamba
- Main problem: strong temporal shift in signal dynamics =  $f(\text{externalities})$ 
  - Use **forecast-correction strategies** with recursive loops to for dynamic updates
  - Try to reverse-engineer physical causalities within the system to reduce noise (ambient conditions, heat transfer,...)